Tom Derrick

Data Scientist · Electricity Markets & Quantitative Analytics · Sydney, Australia · tderrick98@outlook.com · tder311.github.io/energy · github.com/tder311 · linkedin.com/in/tderrick98

PROFILE

Data scientist with four years in Australia's National Electricity Market, across price forecasting, mathematical optimisation, and applied AI. I've worked the market from both sides: inside a generator's trading and risk function at Delta Electricity, and now leading the modelling behind Modo Energy's Australian forecast products. My work is building the models behind the energy sector: MILP dispatch and capacity models solving over the real transmission network, ML price forecasters, contract valuation engines, and RAG LLM systems. I direct the NEM model roadmap, reporting to the Head of Region, and train and delegate to a junior data scientist.

EXPERIENCE

Data Scientist · Modo Energy, Sydney Aug 2025 – present

Series B scaleup building forecasting and investment intelligence for the energy transition. Customers include Goldman Sachs, Apollo, EDF, and Shell. I joined the Australian team in its first year and lead the modelling behind its local products.

- Designed and shipped the nodal transmission network model behind the July 2026 NEM forecast release: DC optimal power flow, bus-level dispatch, PTDF, load disaggregation, transmission augmentation through 2050+. Took the product from regional to 200+ nodes, so assets are valued at the connection point where they actually sit: largest update since the Australian launch and its first on-time release, delivered as sole modeller while training a new data scientist.
- Designed the capacity expansion model (linopy + XPRESS MILP) determining least-cost generation and storage buildout across the NEM, replacing weeks of manual capacity tuning per forecast with a single optimisation.
- Build and run the production cost models producing long-term NEM price forecasts; battery dispatch optimisation and revenue modelling in the multi-market Global Dispatch Model (Pyomo + XPRESS).
- Built the contracting strategy system end to end: a valuation engine pricing PPAs, tolls and offtake structures against forecast dispatch, plus the React configuration UX shipped to customers.
- Built and deployed an LLM support agent (Claude + RAG over model documentation, code, and model outputs) answering customer and internal questions across Slack and Intercom. Retrieval runs across heterogeneous sources, with answers grounded in retrieved context rather than free generation.
- Contributed data engineering and semantic table annotation to Modo's customer-facing AI analyst, so the model interprets forecast datasets correctly.
- Act as the forecasting expert on customer calls alongside the sales team, running technical due diligence with prospects evaluating the product, and discovery with existing customers whose requirements feed back into the NEM model roadmap I direct.
- Present each forecast release at Modo's customer workshops and to the wider business, translating model and methodology changes into what they mean for investment and trading decisions.
- Author Modo's published forecast research: model releases, market outlooks, and deep dives.

Market Analyst · Delta Electricity, Sydney Jul 2022 – Jul 2025

NSW electricity generator. Front-office analytics supporting trading, hedging, and portfolio risk.

- Designed whole-of-market NEM simulation models (PLEXOS/Prophet) used for trading, hedging, and business development decisions.
- Built a Monte-Carlo Gross-Margin-at-Risk engine quantifying portfolio earnings risk across price and volume scenarios, used for risk compliance and portfolio optimisation.
- Built an ML-powered spot-price forecasting model supporting short-term trading decisions.
- Developed LGC and ACCU certificate trading strategy across environmental commodity markets.

- Modelled major investment decisions, including the end-of-life extension of Vales Point Power Station.
- Optimised the planned-outage schedule to minimise revenue impact; built the internal live trading-performance dashboard. Completed Delta's 2024 future-leadership program.

Customer Success Manager · TechTarget, Sydney Nov 2020 – Jun 2022

- Managed APAC campaign delivery for marquee accounts (Red Hat, Dell, IBM, Cisco, Alibaba).

PERSONAL PROJECTS

- **gridline**. Energy-market simulator built to stand in for PLEXOS, visual-first and agent-native: chronological LP dispatch (copperplate, zonal or nodal DC-OPF) with prices from duals and a “why this price” explain, storage, hydro water values and offer curves; builds the NEM from AEMO's ISP PLEXOS model. Python · linopy · HiGHS · React.
- **nem-market-watch** (nemdash.up.railway.app). Live dashboard for Australia's National Electricity Market: five-minute prices and demand, market metrics, PASA reserve outlook, bid-band analysis, and a tool-calling LLM agent that answers questions against live data. React · FastAPI · PostgreSQL · Railway.
- **nem-forward** (forward-frontend-production.up.railway.app). XGBoost price forecaster (7-day, half-hourly, P10–P90) benchmarked against AEMO pre-dispatch, LP battery dispatch optimiser, and bid-band generation; every forecast persisted with its model's training timestamp so past runs are scored against actuals instead of overwritten. XGBoost · FastAPI · React.
- **nem-ingestion**. Five-minute AEMO market data pipeline: six NEMWEB clients, sole writer and schema owner across 13 tables, backfill CLIs, hourly rollups, and a retention sweep that cut the database from 21 GB to 2 GB. Python · PostgreSQL · Railway.
- **pocket-etrm**. Local-first ETRM phone app for a NEM trading desk: futures, bespoke OTC shapes and LGC/ACCU certificates, with positions, MtM, P&L, exposure, limits and parametric VaR marked against live ASX settlements. TypeScript · Expo · SQLite.
- **agent-tui** (github.com/tder311/agent-tui). Terminal UI for local AI coding agents: git worktrees, branches, Claude Code / OpenCode sessions, live running agents, and open PRs, grouped by project via app-agnostic filesystem discovery; read-only by default. Go · Bubble Tea.
- **opencode-orchestrator** (github.com/tder311/opencode-orchestrator · npm). An opencode plugin for parallel AI coding agents: each spawns into an isolated git worktree with its own interactive terminal session, pushing completion/blocker reports back to a supervising session.
- **railway-tui** (github.com/tder311/railway-tui). Terminal dashboard for Railway in Go and Bubble Tea: single static binary, OAuth (PKCE) login, live view of deployments, log streams, and environment variables over Railway's GraphQL API, with restart/redeploy/rollback actions and rate-limit-aware polling.
- **work-graph**. Git-backed dependency tracker for work shared by people and AI agents: CLI, web and terminal UI over one Python core, with per-task git merges so two machines can edit at once. Python · Textual.

EDUCATION & ACCREDITATION

- AFMA Financial Markets Accreditation
- UNSW: Master of Data Science (with Excellence)
- University of Sydney: BSc, Computer Science & Statistics

SKILLS

Languages: Python · SQL · R · SAS · Go · JavaScript

Technical: NEM market modelling · PLEXOS · linopy · Pyomo · XPRESS (MILP/LP) · XGBoost · time-series forecasting · Monte-Carlo simulation & risk modelling · RAG & LLM systems (Claude, OpenAI) · AWS (Batch, S3) · FastAPI · React · PostgreSQL · git

Soft: Client-facing technical due diligence · presenting to technical and non-technical audiences · mentoring and delegating to junior analysts · roadmap ownership and prioritisation · published technical writing